Polynomial End Behavior, Zeros & Multiplicity

End behavior of polynomial functions

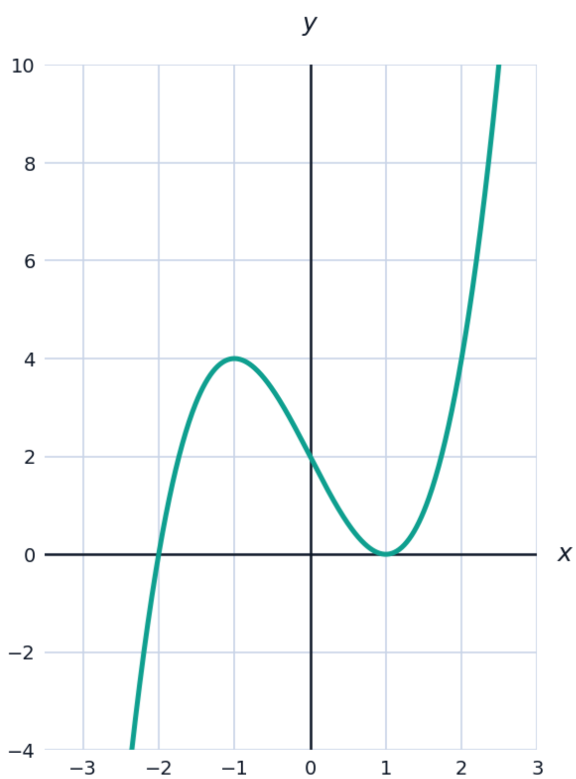
1 State the end behavior of $f(x) = -2x^5 + 3x^2 - 1$, using the leading term.

2 State the end behavior of $f(x) = 3x^4 - 5x^3 + 2$.

Zeros and their multiplicities

3 Find the zeros of $f(x) = (x-2)^3(x+1)^2$ and state the multiplicity and graphical behavior at each.

4 The graph shows $f(x) = (x+2)(x-1)^2$, illustrating a crossing zero at $x = -2$ and a touching zero at $x = 1$.



The Rational Zero Theorem

5 List all possible rational zeros of $f(x) = 2x^3 - 3x^2 - 8x + 12$ using the Rational Zero Theorem, then find the actual zeros.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Constructing a polynomial from its zeros

6 Write a polynomial function of least degree with real coefficients, zeros at $x = -1$ (multiplicity 2) and $x = 3$ (multiplicity 1), and a leading coefficient of 1.

7 A polynomial has zeros at $x = 0$, $x = 2$, and $x = -3$, all with multiplicity 1, and passes through $(1, 8)$. Find the polynomial in the form $f(x) = a \cdot x(x-2)(x+3)$.

Local extrema and turning points

8 A polynomial of degree n has at most how many turning points (local extrema)? State the rule and apply it to a degree-5 polynomial.

9 The graph of a polynomial shows exactly 3 turning points. What is the minimum possible degree of the polynomial?

Average rate of change and concavity

10 Find the average rate of change (AROC) of $f(x) = x^3 - 4x$ on $[1, 3]$.

11 For $f(x) = x^3 - 4x$, compare the AROC on $[0, 1]$ and $[1, 2]$, and use the comparison to determine whether f is concave up or concave down on this interval.

Sign charts and interval analysis

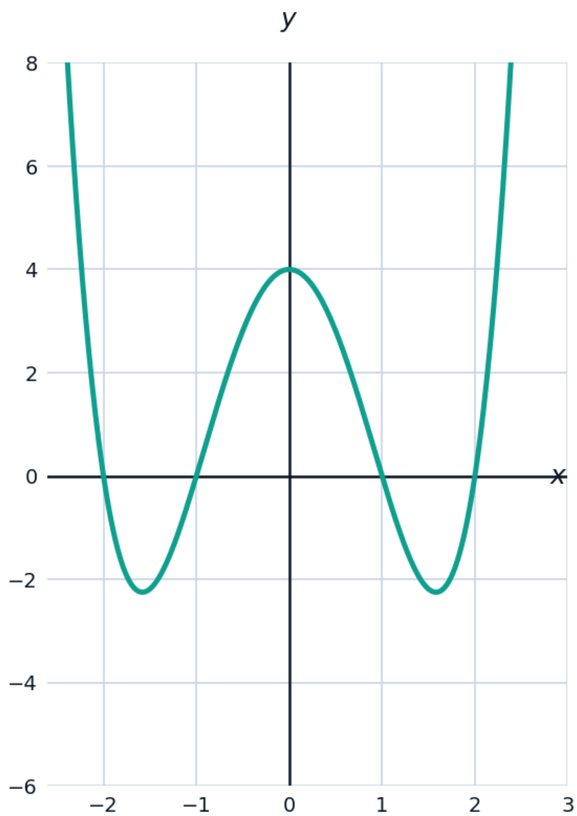
12 Build a sign chart for $f(x) = (x+1)(x-2)(x-4)$ and state the intervals where $f(x) > 0$.

13 Use a sign chart to solve the inequality $(x-1)^2(x+3) \le 0$.

Visualizing end behavior families

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

14 The graph shows $f(x) = x^4 - 5x^2 + 4$, a degree-4 polynomial with four real zeros.



a State the end behavior of the graph as $x \to \pm\infty$.

b Identify the four zeros shown and state whether each is a crossing or touching zero.

c How many turning points does the graph display, and is this consistent with the degree?

Putting it together: a full analysis

15 For $f(x) = x^3 + 2x^2 - 5x - 6$: verify that $x = -1$ is a zero, factor completely, then describe the end behavior and the behavior at each zero.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647